Everyday Science

A Short Study Guide

Three chapters on the water cycle, energy and motion, and the human body, each with review questions and an answer key.

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Chapter 1

The Water Cycle

Water on the Move

Earth has about the same amount of water today as it did millions of years ago. That water is always moving between the oceans, the air, and the land. This endless journey is called the water cycle, and it is powered by heat from the Sun.

Evaporation

When the Sun warms oceans, lakes, and rivers, some of the liquid water turns into an invisible gas called water vapor. This change is called evaporation. Plants also release water vapor through tiny openings in their leaves, a process called transpiration.

Condensation

Warm, moist air rises. As it rises, the air pressure around it drops, so the air expands and cools. Cool air cannot hold as much water vapor, so the vapor turns back into tiny liquid droplets. This is condensation, and billions of these droplets together form clouds.

Precipitation

When cloud droplets join together and become too heavy to stay in the air, they fall to the ground as precipitation. Precipitation can be rain, snow, sleet, or hail, depending on the temperature of the air it falls through.

Collection

Most precipitation lands in the oceans, which hold almost all of the water on Earth. Water that falls on land may flow over the surface as runoff into streams and rivers, or it may soak into the ground. Water stored in spaces within soil and rock is called groundwater. Sooner or later, the Sun warms this water again and the cycle continues.

Review Questions

1. What is the main source of energy that drives the water cycle?
    A. The Moon's gravity
    B. The Sun
    C. Earth's core
    D. Ocean currents

2. Which process turns liquid water into water vapor?
    A. Condensation
    B. Precipitation
    C. Evaporation
    D. Infiltration

3. Clouds form mainly through which process?
    A. Condensation
    B. Transpiration
    C. Runoff
    D. Melting

4. What is transpiration?
    A. Water flowing over the ground into rivers
    B. Water vapor released by plants through their leaves
    C. Ice turning directly into vapor
    D. Rain soaking into the soil

5. Which of these is a form of precipitation?
    A. Fog
    B. Dew
    C. Hail
    D. Steam

6. Where is most of Earth's water stored?
    A. In glaciers
    B. In the oceans
    C. Underground
    D. In the atmosphere

7. Water that soaks into the ground and fills spaces in soil and rock is called what?
    A. Groundwater
    B. Runoff
    C. Humidity
    D. Vapor

8. Why does air cool as it rises?
    A. It gains more water
    B. It moves closer to the Sun
    C. Air pressure drops, so the air expands and cools
    D. It mixes with clouds

Chapter 2

Energy and Motion

What Is Energy?

Energy is the ability to do work or cause change. It comes in many forms, including heat, light, sound, electrical energy, and chemical energy stored in food and fuel. Two of the most important forms for understanding motion are kinetic energy and potential energy.

Kinetic and Potential Energy

Kinetic energy is the energy of motion. A rolling ball, a flowing river, and a flying bird all have kinetic energy. Potential energy is stored energy. A book on a high shelf has gravitational potential energy because of its height. When the book falls, that stored energy turns into kinetic energy.

Conservation of Energy

Energy is never created or destroyed. It only changes from one form to another. This rule is called the law of conservation of energy. When you rub your hands together, the energy of motion turns into heat.

Forces and Friction

A force is a push or a pull. Gravity is a force that pulls objects toward Earth. Friction is a force that resists motion between two surfaces that touch. Friction is why a sliding box slows down and stops.

Newton's Laws of Motion

Isaac Newton described three laws of motion. The first law says an object at rest stays at rest, and an object in motion keeps moving, unless an outside force acts on it. This tendency is called inertia. The second law says a stronger force causes a greater change in motion. The third law says that for every action there is an equal and opposite reaction. Speed describes how far an object travels in a given amount of time.

Review Questions

1. Kinetic energy is the energy of what?
    A. Position
    B. Motion
    C. Heat only
    D. Light only

2. A book sitting on a high shelf has which type of energy because of its height?
    A. Kinetic energy
    B. Sound energy
    C. Gravitational potential energy
    D. Electrical energy

3. According to the law of conservation of energy, what happens to energy?
    A. Machines can create it
    B. Friction destroys it
    C. It changes form but is never created or destroyed
    D. It only exists as heat

4. What is friction?
    A. A force that resists motion between surfaces that touch
    B. A type of stored energy
    C. The speed of an object
    D. A force that pulls objects toward Earth

5. An object at rest stays at rest unless acted on by an outside force. Which law describes this?
    A. Newton's first law
    B. Newton's third law
    C. The law of gravity
    D. The law of conservation of energy

6. What does speed tell you?
    A. How heavy an object is
    B. How far an object travels in a given amount of time
    C. Which direction an object faces
    D. How much energy is stored

7. When you push on a wall, the wall pushes back on you with equal force. Which law is this?
   A. Newton's first law
   B. Newton's second law
   C. Newton's third law
   D. The law of inertia

8. Which change happens when a ball rolls downhill?
   A. Kinetic energy turns into potential energy
   B. Potential energy turns into kinetic energy
   C. All of its energy disappears
   D. The ball gains weight

Chapter 3

The Human Body

Body Systems

The human body is made of systems that work together. Each system is a group of organs with a shared job. Three of the most important are the circulatory system, the respiratory system, and the digestive system. The nervous system, made of the brain, spinal cord, and nerves, controls them all.

The Circulatory System

The heart is a muscle that pumps blood through blood vessels to every part of the body. Red blood cells carry oxygen from the lungs to the rest of the body. White blood cells fight infection. Platelets help blood clot when you get a cut.

The Respiratory System

When you breathe in, a dome-shaped muscle called the diaphragm moves down and pulls air into the lungs. Inside the lungs are millions of tiny air sacs called alveoli. Oxygen passes from the alveoli into the blood, and carbon dioxide passes from the blood into the alveoli to be breathed out.

The Digestive System

Digestion breaks food into nutrients the body can use. Food is broken down in the mouth and stomach, and most nutrients are absorbed into the blood in the small intestine. The large intestine absorbs water from what is left.

Muscles and Bones

The skeleton supports the body and protects organs such as the brain and heart. Muscles are attached to bones. Muscles can only pull, not push, so they move bones by contracting. Many muscles work in pairs, with one pulling a bone one way and the other pulling it back.

Review Questions

1. Which organ pumps blood through the body?
    A. Lungs
    B. Liver
    C. Heart
    D. Stomach

2. Where does gas exchange take place in the lungs?
    A. In the alveoli
    B. In the trachea
    C. In the diaphragm
    D. In the throat

3. Which blood cells carry oxygen?
    A. White blood cells
    B. Platelets
    C. Red blood cells
    D. Plasma

4. What is the main job of the small intestine?
    A. Pumping blood
    B. Absorbing nutrients from food
    C. Filtering air
    D. Making red blood cells

5. Which body system includes the brain, spinal cord, and nerves?
    A. Digestive system
    B. Skeletal system
    C. Nervous system
    D. Respiratory system

6. What do white blood cells do?
    A. Carry oxygen
    B. Fight infection
    C. Digest food
    D. Build bone

7. How do muscles move bones?
    A. By pushing them
    B. By contracting and pulling on them
    C. By dissolving them
    D. By heating them

8. Which muscle helps you breathe by moving down to pull air into the lungs?
    A. The heart
    B. The biceps
    C. The diaphragm
    D. The stomach

Appendix A: Answers to Review Questions

Chapter 1: The Water Cycle

1. B. The Sun heats water at the surface, which causes evaporation and keeps the cycle moving.

2. C. Evaporation happens when heat changes liquid water into a gas.

3. A. As moist air cools, water vapor condenses into tiny droplets that form clouds.

4. B. Plants give off water vapor through small openings in their leaves.

5. C. Hail is ice that falls from clouds. Fog, dew, and steam do not fall from clouds.

6. B. The oceans hold almost all of the water on Earth.

7. A. Groundwater is water stored below the surface in soil and rock.

8. C. Lower pressure higher up lets rising air expand, and expanding air cools.

Chapter 2: Energy and Motion

1. B. Any moving object has kinetic energy.

2. C. Height gives an object stored energy that is released if it falls.

3. C. Energy can change from one form to another, but the total amount stays the same.

4. A. Friction acts where surfaces rub together and slows motion down.

5. A. Newton's first law describes inertia, the tendency of objects to keep doing what they are doing.

6. B. Speed is distance traveled divided by the time it takes.

7. C. Newton's third law says every action has an equal and opposite reaction.

8. B. As the ball loses height, its stored energy becomes energy of motion.

Chapter 3: The Human Body

1. C. The heart is a muscle that pumps blood through the blood vessels.

2. A. Oxygen and carbon dioxide pass between the air and the blood in the tiny air sacs called alveoli.

3. C. Red blood cells pick up oxygen in the lungs and carry it around the body.

4. B. Most nutrients pass into the blood through the walls of the small intestine.

5. C. The nervous system carries signals between the brain and the rest of the body.

6. B. White blood cells defend the body against germs.

7. B. Muscles can only pull, so they move bones by contracting.

8. C. The diaphragm moves down when you breathe in, which draws air into the lungs.